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Generative Adversarial Network for Modeling of CO_2 Plume Evolution in Geological Carbon Storage Systems

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Abstract

Deep learning-based surrogate models offer a powerful alternative to traditional numerical simulations for tackling subsurface multiphase flow challenges, such as those in Geological Carbon Storage (GCS). In this work, we utilized a deep learning surrogate model using Generative Adversarial Networks (GANs) to model CO_2 saturation and pressure buildup evolution. GANs, with their generator-discriminator framework, excel at capturing intricate spatial and temporal patterns, enabling accurate predictions of CO_2 plume saturation and pressure buildup in saline aquifers. By leveraging adversarial training, GANs effectively model structured data, preserving spatial relationships and generating high-resolution outputs. We first developed physics-based numerical simulation models to represent both the injection and post-injection phases of GCS. Using Latin-Hypercube sampling, we created a diverse set of reservoir and decision parameters, forming a comprehensive simulation database. The GANs were trained and evaluated on a 2D model to assess their performance. During training, we employed Mean Squared Error (MSE) and spatial derivative-based loss functions to optimize the model hyperparameters. The GANs demonstrated strong performance, achieved R^2 values of 0.989 and 0.996 for saturation and pressure buildup predictions, respectively. The Normalized Absolute Error (NAE) remained consistently around 1% across all predictions, highlighting the model's accuracy in capturing the temporal and spatial evolution of CO_2 saturation. Additionally, the GANs demonstrated remarkable computational efficiency, with prediction times of just 0.01 seconds per case, compared to 1000 seconds for the 2D model using physics-based simulations. These results underscore the potential of GANs to deliver highly accurate predictions while significantly reducing computational costs.

Keywords: Reservoir Simulation, Geological Carbon Storage, GANs, Generative AI, Rapid Predictions

Introduction

Climate change, driven by anthropogenic greenhouse gas emissions, poses a significant threat to our planet's ecosystems and human societies (Ali et al., 2025; Izgec et al., 2006; Kim et al., 2010; Le Gallo et al., 2002;

Mohamed et al., 2010). Carbon dioxide (CO_2), the primary contributor to global warming, necessitates urgent mitigation strategies to reduce its atmospheric concentration and curb its impacts (Aslam et al., 2024; Khamidy et al., 2019). Among the emerging solutions, Geological Carbon Storage (GCS) has gained prominence as a viable technology for combating climate change. GCS involves capturing CO_2 from industrial sources or directly from the atmosphere, compressing it into a supercritical state, and injecting it into deep geological formations such as saline aquifers or depleted oil and gas reservoirs (Fatima et al., 2021). These formations provide secure, long-term storage through a combination of trapping mechanisms, including physical trapping within porous rock, dissolution in formation brine, and mineral trapping via chemical reactions that convert CO_2 into stable carbonate minerals. As a negative emissions technology, GCS not only helps reduce existing atmospheric CO_2 but also enables the continued use of fossil fuels when coupled with carbon capture technologies. However, the large-scale deployment of GCS requires extensive research and development to ensure its safety, efficiency, and cost-effectiveness (Pan et al., 2019; Yao et al., 2022).

Numerical modeling plays a critical role in understanding and predicting the behavior of CO_2 in subsurface environments, making it an indispensable tool for GCS (Lengler et al., 2010; Sasaki et al., 2008). These models integrate a wide range of physical and chemical processes, including multiphase fluid flow, heat transfer, and geochemical reactions, to simulate the complex interactions between injected CO_2, resident fluids, and the rock matrix (Grigg and Svec, 2006; Nguyen, 2003). Subsurface environments are highly heterogeneous, characterized by complex geological formations, varying porosities, permeabilities, and fluid properties, which makes accurately modeling CO_2 behavior a challenging yet crucial task. The success of GCS projects heavily depends on the ability to predict CO_2 plume migration, pressure buildup, and the associated risks of leakage and capillary trapping. However, high-fidelity numerical simulations that incorporate these complexities are computationally intensive and time-consuming, often requiring days to weeks for a single simulation run (Yan et al., 2023b). This extended computation time is largely due to the need to account for large spatial domains, complex boundary conditions, and long simulation timeframes, all of which involve solving intricate partial differential equations.

Such computational demands hinder real-time decision-making and limit the ability to perform extensive scenario analyses, which are essential for optimizing GCS operations (Feng et al., 2025; Tariq et al., 2025). These limitations are particularly apparent in scenarios requiring high-resolution simulations or those involving large-scale reservoir models (Tariq et al., 2024). While numerical models provide high accuracy, the slow speed of computation can restrict their utility in dynamic, real-time decision-making contexts, such as monitoring CO_2 injection and plume migration during an active GCS project. Consequently, there is a need for alternative approaches that can deliver rapid predictions without sacrificing the accuracy required for reliable decision support. To address these challenges, in recent years, deep learning-based surrogate models have emerged as a promising alternative, offering significant reductions in computation time while maintaining acceptable levels of accuracy (Tariq et al., 2023a, 2023c). These models, particularly neural operators, can be trained on large datasets generated from high-fidelity simulations or field data to learn the underlying patterns and relationships governing CO_2 behavior in the subsurface (Feng et al., 2024). Once trained, surrogate models can provide near-instantaneous predictions, enabling rapid decision-making and scenario analysis (Wen et al., 2022). This capability is especially valuable for applications that require fast iteration, such as optimization of injections and monitoring strategies in GCS. Figure 1 compares the advantages and limitations of high-fidelity numerical simulators with deep learning-based proxy models.

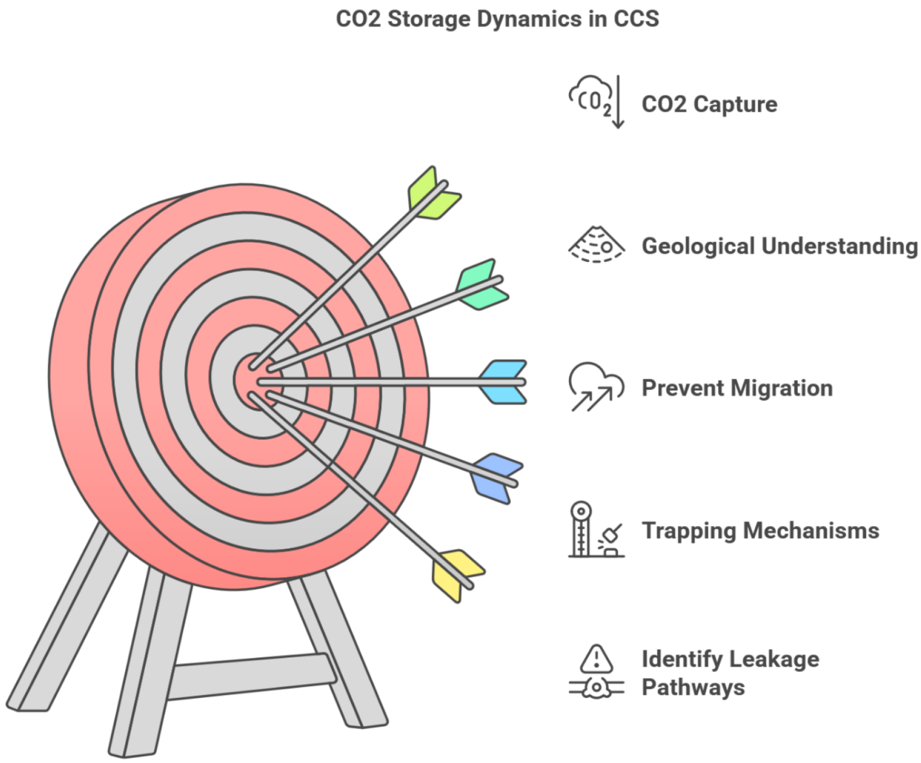


Figure 1—Challenges of geological carbon storage in saline aquifers.

Traditional reservoir simulations are computationally expensive, particularly when modeling complex geological formations over long time periods (Feng et al., 2024; Tariq et al., 2025, 2023a, 2023b, 2023c; Yan et al., 2023a). While surrogate models offer a faster alternative, they can still be resource-intensive to train. To address this challenge, we propose the use of Generative Adversarial Networks (GANs) due to their speed, accuracy, and scalability. By integrating GANs with high-fidelity simulation data, we aim to strike a balance between computational efficiency and predictive performance. Figure 2 highlights the comparative advantages and limitations of high-fidelity simulations and deep learning-based surrogate models.

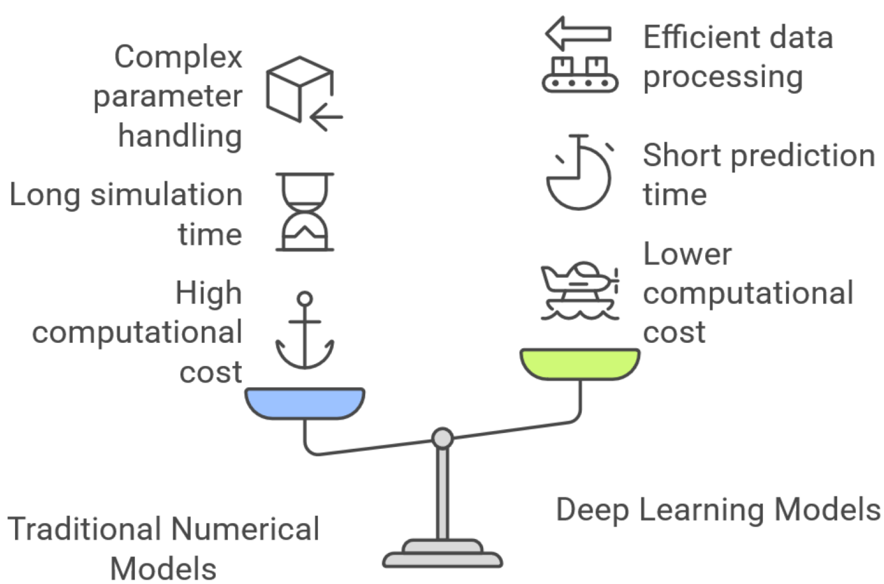


Figure 2—Comparison of high-fidelity simulations and deep learning techniques.

GANs, originally developed for generating realistic synthetic data, have recently emerged as powerful tools for modeling complex physical systems, including subsurface flow dynamics (Aldausari et al., 2023; Goodfellow et al., 2020; Gui et al., 2023; Li et al., 2022; Nguyen et al., 2022; Zhang et al., 2023). A GAN comprises two neural networks—a generator and a discriminator—that are trained in tandem through an adversarial process. The generator produces synthetic outputs intended to resemble real data, while the discriminator evaluates the authenticity of these outputs. As training progresses, the generator becomes increasingly proficient at producing realistic data distributions, and the discriminator becomes better at distinguishing real from synthetic data. This adversarial learning mechanism enables GANs to generate high-resolution, data-consistent outputs that accurately capture the spatiotemporal behavior of physical systems.

Because of their ability to process large datasets and generate high-quality outputs efficiently, GANs are well suited for surrogate modeling in geological carbon storage (GCS). Once trained, GAN-based models can rapidly predict key variables such as CO_2 saturation and reservoir pressure, enabling real-time monitoring, scenario analysis, and operational optimization. This significantly accelerates the simulation workflow while preserving essential physical realism, making GANs particularly valuable for large-scale GCS projects.

In this study, we develop a GAN-based surrogate model to predict CO_2 plume migration in heterogeneous saline reservoirs. The model enables simultaneous estimation of pressure buildup and gas saturation distributions across all time steps. By leveraging parallel computations, it dramatically reduces prediction time compared to conventional numerical simulators. Trained on high-fidelity simulation data, the model accurately captures the intricate spatial and temporal dynamics of CO_2 behavior, making it well suited for real-time decision-making and uncertainty quantification.

This approach bridges the gap between computationally intensive high-fidelity reservoir simulations and the need for rapid predictions in practical applications. As a result, it supports the efficient and safe deployment of GCS technologies as part of global climate change mitigation strategies. The remainder of this paper is organized as follows: Section 2 introduces the governing equations of multiphase flow, the architecture of the proposed GAN-based surrogate model, and the optimization algorithm used for training. Section 3 presents the model's prediction results and evaluates its performance against traditional simulation approaches. Finally, Section 4 concludes with a summary of key findings and outlines potential directions for future research on GAN-based surrogate modeling in the context of GCS.

Computational Methodology

Physical Model Description

In this study, we utilized a commercial numerical simulator (CMG, 2022) to model the multiphase flow of CO_2 and brine within porous media. The simulation model leverages the finite difference scheme to discretize the partial differential equations for mass conservations. Fluid properties are determined using the Peng Robinson equation of state (EOS). The governing equation for transport and flow is given by Eq. (1),

$$\frac{\partial}{\partial t}\left(\varphi \sum_{\alpha} S_{\alpha}\rho_{\alpha}x_{\alpha,i}\right) - \nabla \cdot \left\{K \sum_{\alpha} \frac{k_{r\alpha}}{\mu_{\alpha}} \rho_{\alpha}x_{\alpha,i}\left(\nabla p_{\alpha} + \rho_{\alpha} g \nabla Z\right)\right\} + \sum_{1}\left(\sum_{\alpha} \rho_{\alpha}x_{\alpha,i}q_{\alpha}\right)^{1} = 0 \tag{1}$$

The above equation governs multiphase fluid flow and associated chemical reactions within a reservoir. It incorporates three key terms: the first representing the rate of change of fluid mass, the second capturing the advective transport of fluids based on Darcy's law, and the third term accounting for the presence of sources or sinks. The equation utilizes various symbols to represent critical reservoir properties. These include φ for porosity, t for time, S_α for the saturation of a specific fluid phase (α), ρ_α for the density of a specific fluid phase (α), $x_{\alpha,i}$ for the mole fraction of component i in fluid phase α, k for the rock intrinsic permeability, $k_{r\alpha}$

for the relative permeability of a specific fluid phase (α), μ_α for the viscosity of a specific fluid phase (α), p for fluid pressure, g for gravity, z for depth, and q_α for the well production or injection rate for a specific fluid phase (α) through well perforation l. Subscripts differentiate between primary fluid components (water and CO_2) and distinct fluid phases (supercritical CO_2 and brine).

In the GCS project, deep saline aquifers were injected with supercritical phase CO_2 after being initially saturated with saline water. Hence, the two-phase relative permeability for aqueous and supercritical phases becomes vital in determining their effective phase permeability. This study employs the (Brooks and Corey, 1964) correlation which is given by Eqs. (2 – 5).

$$k_{rw} = \overline{S}_w^{(3+2/\lambda)} \tag{2}$$

$$k_{rg} = \overline{S_g}\left(1-\left(1-\overline{S_g}\right)^{(1+2/\lambda)}\right) \tag{3}$$

$$\overline{S_w} = \frac{S_w - S_{wi}}{1 - S_{gi} - S_{wi}} \tag{4}$$

$$\overline{S_g} = \frac{S_g - S_{gi}}{1 - S_{gi} - S_{wi}} \tag{5}$$

where the variable λ represents the index for pore size distribution, k_{rw} refers to the relative permeability of the brine, k_{rg} denotes the relative permeability of the supercritical CO_2, S_{wi} refers to irreducible brine saturation, S_{gi} indicates the irreducible CO_2 saturation. Peng Robinson EOS was used to estimate the density of the CO_2 while Jossi correlation was used to compute the viscosity of CO_2 (Jossi et al., 1962).

Figure 3 shows the conceptual reservoir model used in this study. Supercritical CO_2 injection occurred through a centrally located vertical well with a radius of 0.076 m at a constant rate ranging from 0.2 to 2.0 mt/year for 30 years. The injection zone was perforated across a specific depth interval. Following the injection phase, CO_2 plume migration was monitored for an additional 170 years. The thickness of the reservoir varies between 25 to 150 meters, with no-flow boundaries on the top and bottom. The outer infinite acting boundary is 10000 meters from the central injection well. This configuration will enable detailed analysis of flow dynamics, pressure distribution, and fluid migration patterns within the reservoir.

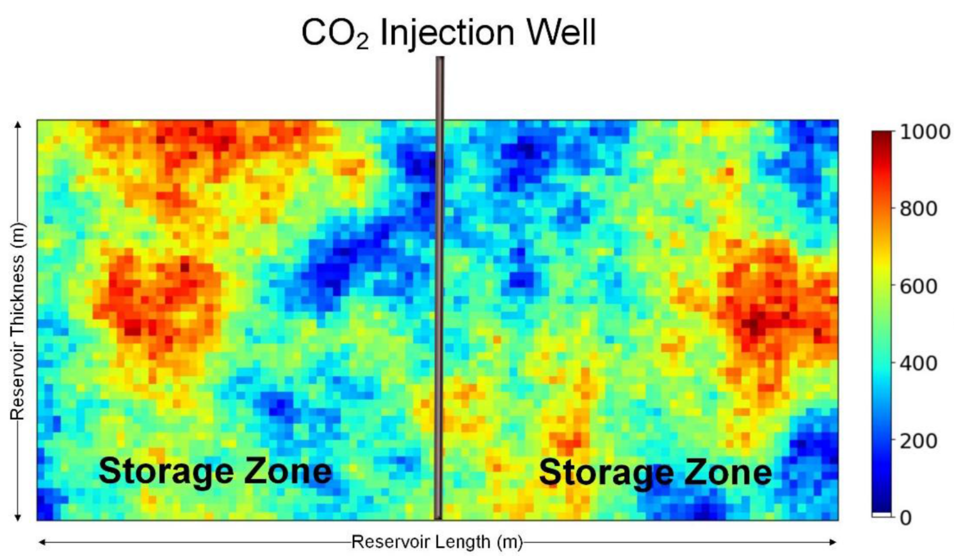


Figure 3—The image depicts a conceptual model view of the reservoir in two dimensions. The permeability distribution map is represented by a color bar in mili-darcy (mD).

Model Workflow

The DL model developed in this study aims to predict the spatial and temporal evolution of CO_2 pressure buildup and CO_2 saturation plume within the storage reservoir. The training data for the GANs model is derived from high-fidelity reservoir simulations conducted using a state-of-the-art CO_2 storage simulator. Figure 4 the deep learning workflow.

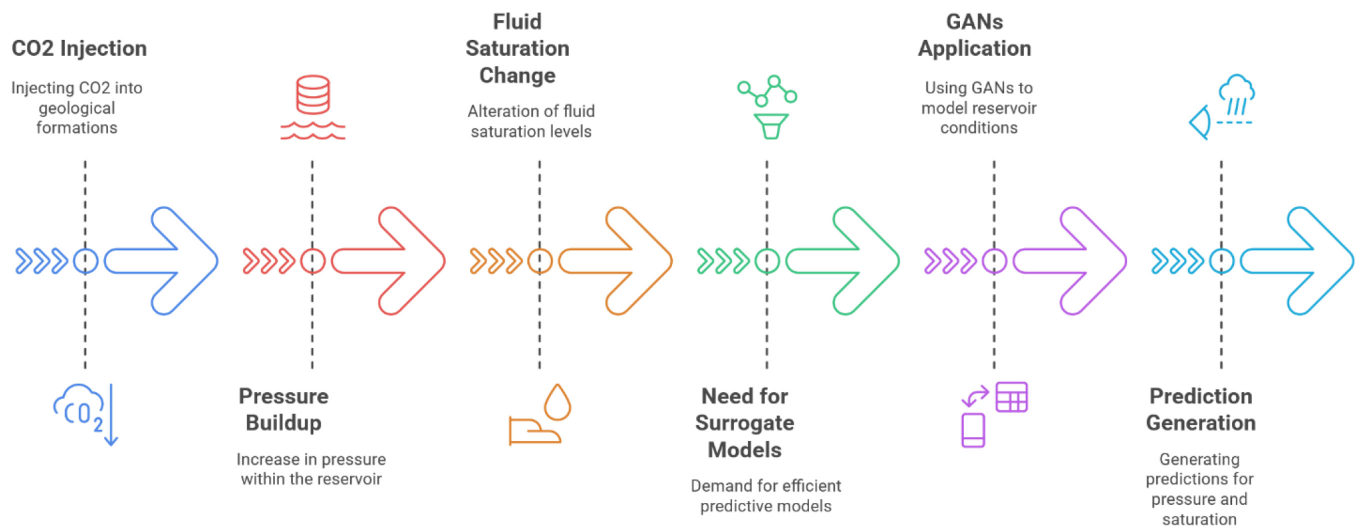


Figure 4—Deep learning workflow adopted in this study.

The details of the input parameters and their data type used to predict the CO_2 pressure buildup and CO_2 saturation plume are summarized in Tables 1. These parameters include reservoir properties like porosity, permeability, pore volume, salinity, reservoir temperature, and reservoir thickness. In addition, operational parameters like injection rates, perforation thickness, and time steps were also considered. We merged the cumulative injected CO_2 volume and the well perforation thickness into one feature map for (Number 4 in Table 1). In this integrated feature (cumulative injected CO_2 volume and the well perforation thickness) the non-zero values at the well grids denoting the total injected CO_2 volume till the current time step and zero values denoting the reservoir. The input features, including scalars, time series and spatial data, were broadcast as images for an end-to-end training manner. This way by learning from a large dataset of reservoir simulations, the DL model can make accurate predictions about pressure and CO_2 saturation distribution within the reservoir, facilitating a more efficient and accurate evaluation of potential storage sites. Data preprocessing involves normalization of input features and target variables to ensure efficient model training.

Table 1—Details of input features.

No.	Input features	Data type
1	Reservoir Porosity	Spatial fields
2	Reservoir Permeability	
3	Grid volume	
4	Cumulative injection & perforation thickness	
5	Temperature	Scalars
6	Salinity	
7	Reservoir thickness	
8	Injection rate	
9	Time step	Time series

Generative Adversarial Networks (GANs) Architecture

The GANs consists of two neural networks: the generator and the discriminator. The generator learns to generate data that mimics the real distribution, while the discriminator evaluates the authenticity of the generated data. In the context of this study, the generator is tasked with generating predictions of pressure buildup and saturation, and the discriminator assesses the accuracy of these predictions. The GANs model architecture used in this study consists of a deep neural network for both the generator and the discriminator. The generator takes in reservoir input parameters (e.g., injection rate, permeability, porosity, reservoir pressure, and temperature) and outputs predictions for pressure and saturation. The discriminator receives both real simulation data and generated data and learns to distinguish between them.

The architecture of the Generative Adversarial Network (GAN) employed in this study consists of two primary components: the generator and the discriminator. The generator is responsible for producing synthetic data, in this case, pressure buildup and saturation fields for subsurface modeling. The generator learns to map a latent vector to the desired output space using a series of convolutional and transpose convolutional layers. Mathematically, the generator can be described as a mapping function $: R^z \rightarrow R^x$, where z is the latent vector (noise), and x represents the generated data (pressure and saturation fields).

The generator takes a latent vector $z \in R^Z$ as input and processes it through multiple layers, including encoders and decoders. Each encoder block consists of a convolutional layer with batch normalization and ReLU activation. The convolution operation in the first encoder layer is expressed as:

$$H_1 = \mathrm{ReLU}(\mathrm{BatchNorm}(\mathrm{Conv}(z; W_1, \quad b_1))) \tag{6}$$

where H_1 represents the feature map after applying the convolution operation with weights W_1 and biases b_1, followed by batch normalization and ReLU activation. The convolution operation reduces the spatial resolution of the input data while increasing the number of filters. This process is repeated in subsequent encoder blocks, with each block having progressively larger numbers of filters. The spatial dimensions of the feature maps are further reduced through pooling layers.

The bottleneck layer, at the end of the encoder, captures the most abstract and compressed features. This layer can be represented as:

$$\mathrm{H_{bottleneck}} = \mathrm{ReLU}(\mathrm{BatchNorm}(\mathrm{Conv}(\mathrm{H_{prev}}; \mathrm{W_{bottleneck}}, \mathrm{b_{bottleneck}}))) \tag{7}$$

where H_{prev} is the output from the previous layer. The features are compressed and passed into the decoder part of the generator.

In the decoder, transpose convolution layers are used to upsample the feature maps and restore the original spatial resolution. The upscaling operation can be mathematically defined as:

$$H_{upscaled} = \mathrm{ReLU}(\mathrm{BatchNorm}(\mathrm{ConvTranspose}(\mathrm{H_{bottleneck}}; \mathrm{W}_{DEC}, \quad \mathrm{b}_{DEC}))) \tag{8}$$

where $H_{bottleneck}$ is the input to the decoder, and W_{dec} and b_{dec} are the weights and biases of the transpose convolution operation. This process is repeated until the feature map is restored to the original resolution. The final output layer produces a single channel, representing the synthetic pressure and saturation fields:

$$\hat{x} = G(z; \theta_G) \tag{9}$$

where $\hat{x}$ is the generated output, and θ_G represents all the parameters of the generator network.

The discriminator, on the other hand, is tasked with distinguishing between real and generated data. It takes either real data $x \in R^x$ or synthetic data $\hat{x}$, and outputs a scalar value indicating whether the input is real or fake. The discriminator is a binary classifier that learns to map data to a probability $D(x; \theta_D)$ that the input x is real. The discriminator function D can be expressed as:

$$D(x; \theta_D) = \sigma(\mathrm{Conv}(x; W_D, b_D)) \tag{10}$$

where σ is the sigmoid activation function, W_D and b_D are the weights and biases for the convolutional layer, and θ_D represents the parameters of the discriminator.

The discriminator's goal is to maximize the probability of assigning a real label to real data and a fake label to generate data. The objective function for the GANs is given by the minimax game between the generator and discriminator:

$$min\theta_G max\theta_D E x{\sim}p_{data}\left[log\ D(x;\theta_D)\right]+Ez{\sim}p_z\left[log\left(1-D\left(G(z);\theta_G;\theta_D\right)\right)\right] \tag{11}$$

where p_{data} is the distribution of real data, and p_z is the distribution of latent vectors. The generator tries to minimize *log (1 – D(G(z)* while the discriminator aims to maximize log D(x) + log (1 – D(x̂)), where $\hat{x}$ = G(z) is the generated data.

Through this adversarial training process, the generator learns to produce data that closely approximates real observations, while the discriminator becomes better at distinguishing real from fake data. The generator and discriminator improve simultaneously, with the generator producing increasingly realistic data, and the discriminator improving its ability to detect fakes. This process leads to high-quality synthetic data generation suitable for applications such as subsurface flow modeling. The layer-wise details of the GAN-based model adopted in this study are presented in Table 2, while the overall architecture of the GANs model is illustrated in Figure 5.

Table 2—Details of the GANs model architecture.

Component	Layer	Input Size	Output Size	Kernel Size	Stride	Filters	Activation	Other Details
Generator	**encoder1**	10×50×50	32×32×32	(3,3)	(1,1)	32	ReLU	Conv2d, BatchNorm
	encoder2	32×32×32	16×16×64	(3,3)	(1,1)	64	ReLU	Conv2d, BatchNorm
	encoder3	16×16×64	8×8×128	(3,3)	(1,1)	128	ReLU	Conv2d, BatchNorm
	encoder4	8×8×128	4×4×256	(3,3)	(1,1)	256	ReLU	Conv2d, BatchNorm
	bottleneck	4×4×256	4×4×512	(3,3)	(1,1)	512	ReLU	Conv2d, BatchNorm
	upscaling4	4×4×512	8×8×256	(2,2)	(2,2)	256	-	ConvTranspose2d
	decoder4	8×8×512	8×8×256	(3,3)	(1,1)	256	ReLU	Conv2d, BatchNorm
	upscaling3	8×8×256	16×16×128	(2,2)	(2,2)	128	-	ConvTranspose2d
	decoder3	16×16×256	16×16×128	(3,3)	(1,1)	128	ReLU	Conv2d, BatchNorm
	upscaling2	16×16×128	32×32×64	(2,2)	(2,2)	64	-	ConvTranspose2d
	decoder2	32×32×128	32×32×64	(3,3)	(1,1)	64	ReLU	Conv2d, BatchNorm
	upscaling1	32×32×64	64×64×32	(2,2)	(2,2)	32	-	ConvTranspose2d
	decoder1	64×64×64	64×64×32	(3,3)	(1,1)	32	ReLU	Conv2d, BatchNorm
	conv	64×64×32	64×64×1	(1,1)	(1,1)	1	LeakyReLU (negative slope=0.01)	Conv2d
Discriminator	**main**	11×50×50	2048×1	(4,4)	(2,2)	2048	LeakyReLU (negative slope=0.01)	Conv2d, LeakyReLU
	conv1	2048	1	-	-	-	-	AdaptiveAvgPool2d, Flatten, Linear
Feature ResNet (netF)	**resnet**	3×224×224	1000	(7,7)	(2,2)	64	ReLU	Conv2d, BatchNorm, MaxPool2d, BasicBlock Layers

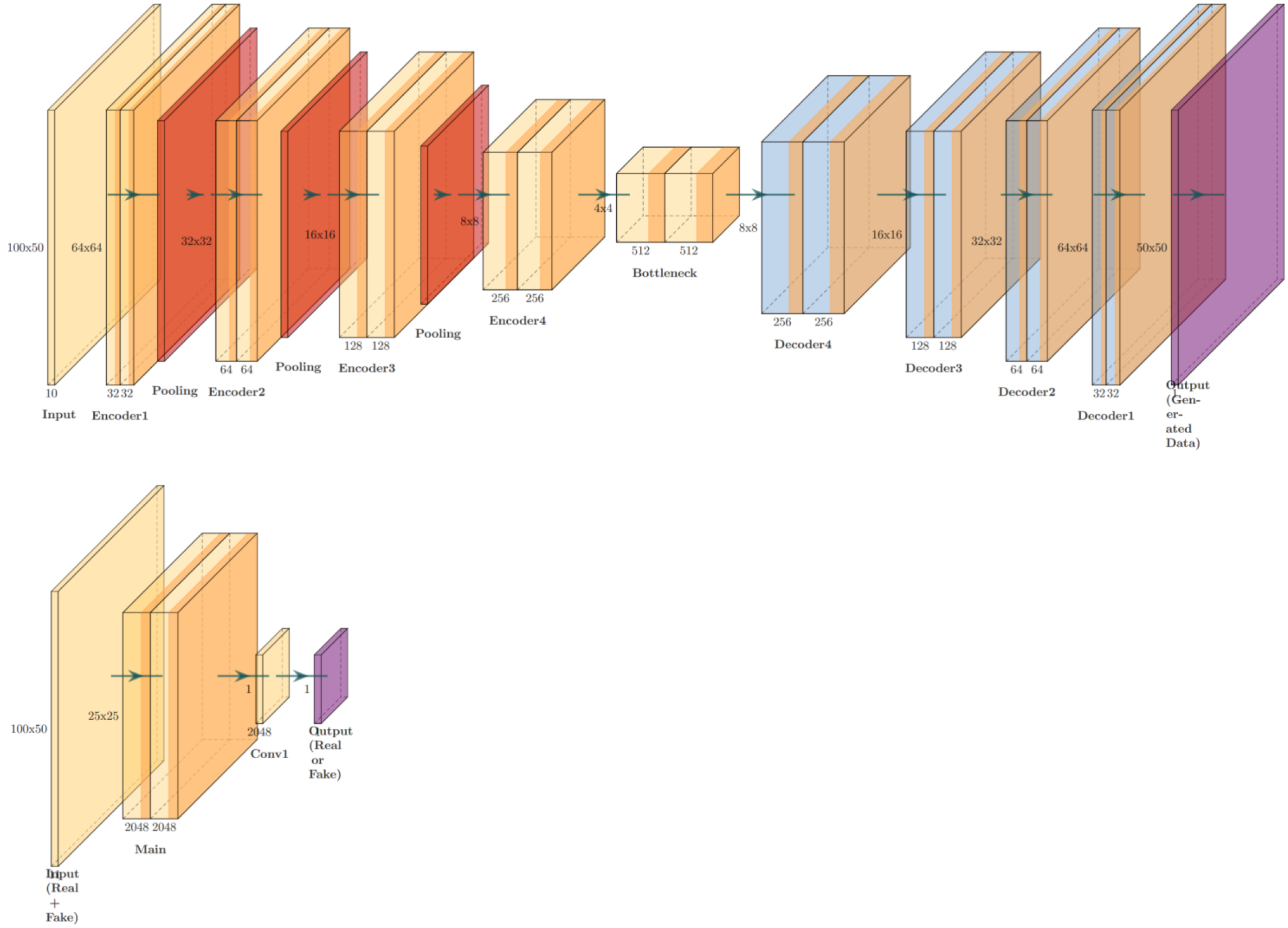


Figure 5—The architecture of the proposed GANS model for GCS parameters prediction.

GANs Training

The GANs model is trained using a dataset split into training and validation sets. The generator is optimized to minimize the difference between predicted pressure and saturation values and the actual values from the reservoir simulations. The discriminator is trained to differentiate between real and generated data, encouraging the generator to produce more accurate predictions. The training process involves iterative updates to both the generator and discriminator using backpropagation.

The proposed model was implemented in the open-source library PyTorch. The loss function used in training combines the adversarial loss, which measures the ability of the discriminator to distinguish between real and generated data, and the mean squared error (MSE) loss, which ensures that the generated predictions are close to the true values.

$$L(Y, \hat{Y}) = \| Y - \hat{Y} \|^2 \tag{12}$$

where Y is the label, $\hat{Y}$ is the prediction.

Eq. 12 is the squared Euclidean distance (or L2 norm) between the true values Y and the predicted values $\hat{Y}$. It penalizes large deviations between the actual and predicted values. We trained the DL model for pressure buildup and saturation respectively for 500 epochs on a NVIDIA RTX A6000 GPU. The initial learning rate was 0.001 that automatically decreases when there was no improvement of the validation performance. The trained model was then evaluated on the test dataset.

To evaluate the performance of the GANS-based surrogate model, we compare its predictions with those of traditional reservoir simulation models. The comparison metrics include the coefficient of determination (R^2), mean absolute percentage error (MAPE), and root mean squared error (RMSE). Additionally, the model's ability to generalize to unseen reservoir conditions is assessed by testing it on a separate test set of simulation data.

Results and Discussions

In this study, we simulated CO_2 injection for 30 years and monitored the reservoir for an additional 170 years. This extended monitoring period is essential to evaluate the long-term stability and containment of the injected CO_2. The reservoir is characterized by a heterogeneous system, which more accurately reflects the natural variability in geological formations. The simulation employs a single injection well, a common configuration in GCS projects. The reservoir is assumed to have an infinite acting boundary, meaning the boundaries do not influence the pressure and flow dynamics within the reservoir over its operational life. A constant injection rate is maintained throughout the 30-year injection period, and all simulations are performed under isothermal conditions. Table 3 presents the numerical simulation input parameters for the base case of both cases.

Table 3—Base case numerical simulation parameters.

Properties	Values	Units
Reservoir	Cartesian System	
Number of Grids	5000	-
Grids in x – y – z directions	100, 1, 50	-
Size of Grid in x direction	Fixed	-
Size of Grid in z directions	50	-
Size of Grid in y	1	
Thickness of Reservoir	100 (base case)	m
Reservoir depth	2400.0	m
Reservoir length	10000	m
Reservoir Length in x-direction	-	m
Reservoir Length in y-direction	-	m
Rock and Fluid Properties		
Reservoir Rock compressibility	3.0×10^{-6}	psi^{-1}
Reservoir Rock Porosity	24.5	%
Permeability Horizontal (kh)	Heterogeneous (0.1 – 1000)	mD
Permeability Vertical	$0.10 \times kh$	mD
Initial Reservoir Pressure	25900.0	kPa
Initial Reservoir Temperature	90	°C
Reservoir Salinity	57100	ppm
Irreducible water saturation, S_{wi}	0.19	-
The density of aquifer water	1021	Kg/m^3
Well, Control and Constraints		
CO_2 Injection Rate	2	mt/year
CO_2 Injection Composition	100% CO_2	
CO2 Injection Well Location	Center	
CO_2 Injection Well Control Mode	Rate	
CO_2 Injection Well Constraint	BHP	psi
Well Diameter	0.370	ft

In this study, the performance of the DL models was evaluated using the coefficient of determination (R^2), root mean square error (RMSE) and structural similarity index measure (SSIM). For the visualization of spatial error distribution, we use the normalized absolute error (NAE), which is the absolute error normalized by the data range at every datapoint. The equations to determine these metrics are shown in Equations 13 – 16.

$$NAE = \frac{\frac{1}{n_{samples}} \sum_{i=1}^{n} \frac{|y_{true} - y_{pred}|}{y_{true}}}{y_{true_{max}} - y_{true_{min}}} \tag{13}$$

$$RMSE = \sqrt{\frac{1}{n_{samples}} \sum_{i=1}^{n} \left(y_{true} - y_{pred}\right)^2} \tag{14}$$

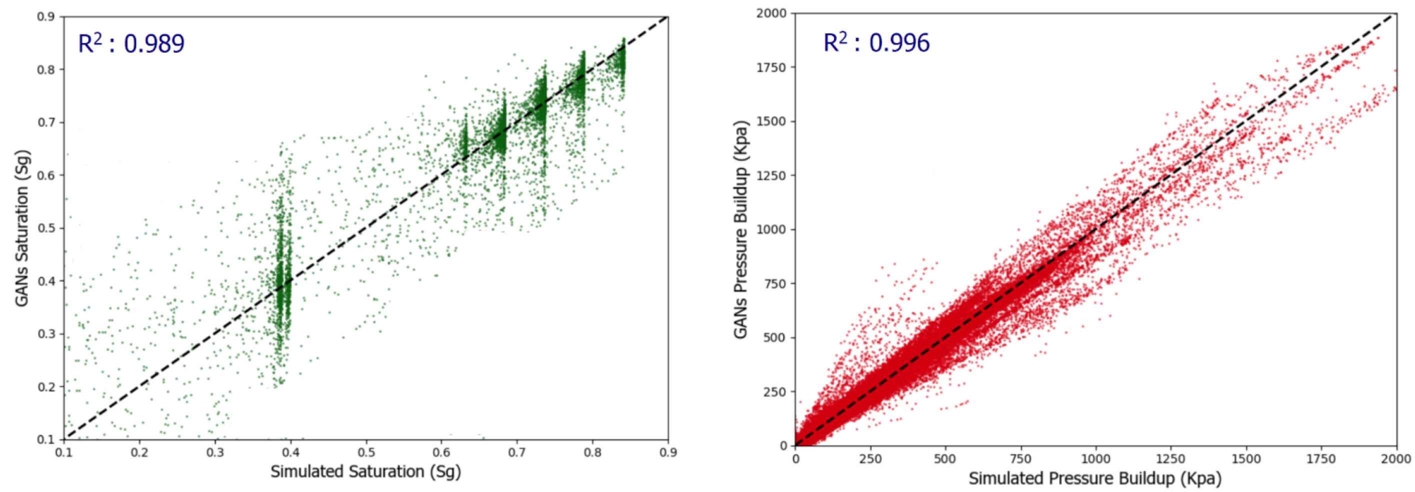
$$R^2 = \left(\frac{\sum_{i=1}^{n} (\boldsymbol{x_i} - \bar{\boldsymbol{x}})(\boldsymbol{y_i} - \bar{\boldsymbol{y}})}{\sqrt{\sum_{i=1}^{n} (\boldsymbol{x_i} - \bar{\boldsymbol{x}})^2} \sqrt{\sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} (y_i - \bar{y})^2}} \right)^2 \tag{15}$$

$$SSIM\,(u, v) = \frac{1}{M} \sum_{m=1}^{M} \frac{\left(2\mu_{u,m}\mu_{v,m} + K_1^2\right)\left(2\sigma_{uv,m} + K_2^2\right)}{\left(\mu_{u,m}^2 + \mu_{v,m}^2 + K_1^2\right)\left(\sigma_{u,m}^2 + \sigma_{v,m}^2 + K_2^2\right)} \tag{16}$$

where y_{true} is the ground truth value, y_{pred} is the model predicted value, $n_{samples}$ is the total data points, *x and y* are the two variables. The *u* and *v* are respectively the true and the predicted values (images). The SSIM is evaluated in *M* local windows in the images of *u* and *v*; μ denotes the mean and σ represented the standard deviation; K_1 and K_2 are small constants to avoid zero values in the denominator terms. The minimum value of SSIM is 0, indicating totally different images, while the maximum value is 1, indicating exactly same images.

The GANS-based surrogate model successfully converges after several iterations, with both the generator and discriminator achieving optimal performance. The generator produces accurate predictions of pressure buildup and CO2 saturation, with low error compared to the reference simulation data. The discriminator effectively distinguishes between real and generated data, indicating that the model has learned the underlying distribution of pressure and saturation in the reservoir.

The trained DL model was evaluated on an unseen test set. Figure 6 shows the overall scatter plot for the testing cases at all the time steps. The R^2 scores for saturation and pressure predictions are 0.989 and 0.996, respectively, demonstrating the satisfactory accuracy of the DL model. Figure 7 shows the testing temporal SSIM, R^2 and RMSE scores.



(a) Saturation (b) Pressure

Figure 6—Scatter plot for (a) saturation and (b) pressure. The color of one datapoint corresponds to the time step.

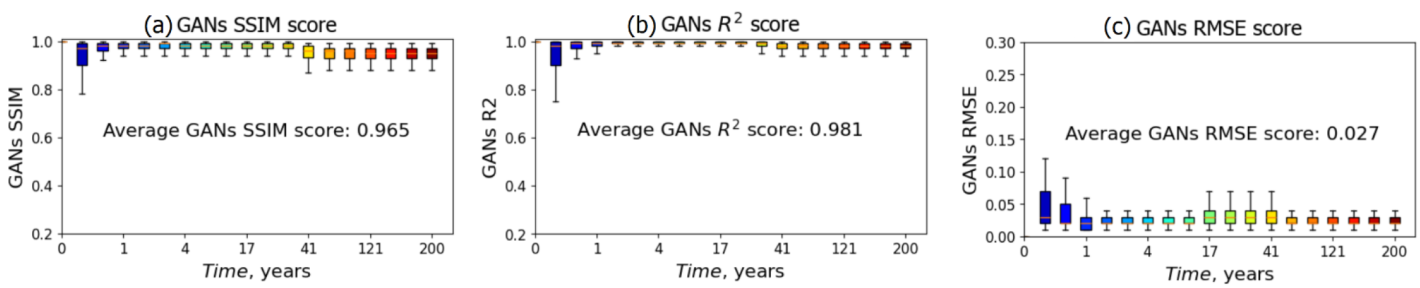


Figure 7—(a) Temporal SSIM, b) Temporal R2 score, and c) Temporal RMSE score

We randomly chose a testing case number 22, whose petrophysical parameters; including permeability, porosity, and cumulative injection perforation are presented in Figure 8. These image data, along with other input features, were used to predict the spatial-temporal evolution of saturation and pressure, respectively.

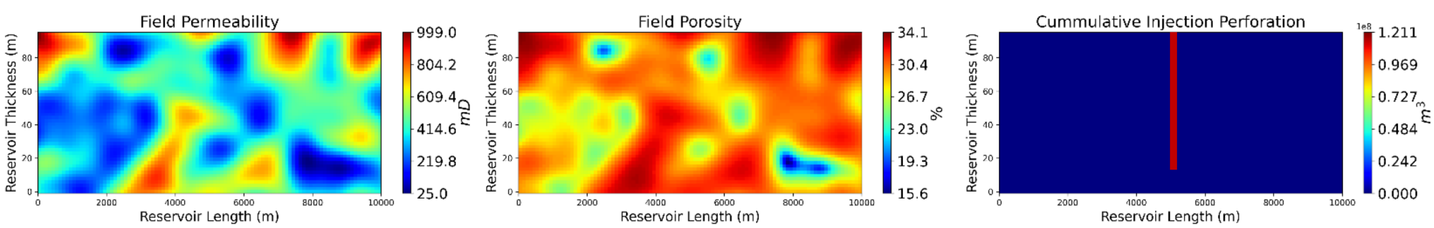


Figure 8—Heterogeneous permeability field, porosity field, well perforation and cumulative injection rates for random testing case number 22. Note the well perforation location has been zoomed in for visualization.

Figure 9 shows the contour visualization of spatial distribution of the saturation field at different time steps. The first column presents the ground truth from the numerical simulator as a reference solution. The second column shows predictions from the GANs model. In addition, the third column is the absolute residual error. Each row shows the values at 1, 2, 10, 20, 30, 100, and 200 years. Note that the first four rows are within the injection period, while the last two rows are data collected from post-injection period. The CO_2 plume initially expands from the well perforation location and moves upward due to buoyancy. During the post-injection period, CO_2 dissolves in saline, leading to a decrease of gas saturation near the plume front. This complex spatial-temporal pattern is well captured by the DL model. Only some minor discrepancy exists around CO_2 plume edge, while the plume morphology is well characterized by the DL model. The overall RMSE is less than 0.02, indicating a close correspondence between the predictions and the numerical simulation results. The average metrics including RMSE, R^2 and SSIM are also satisfactory.

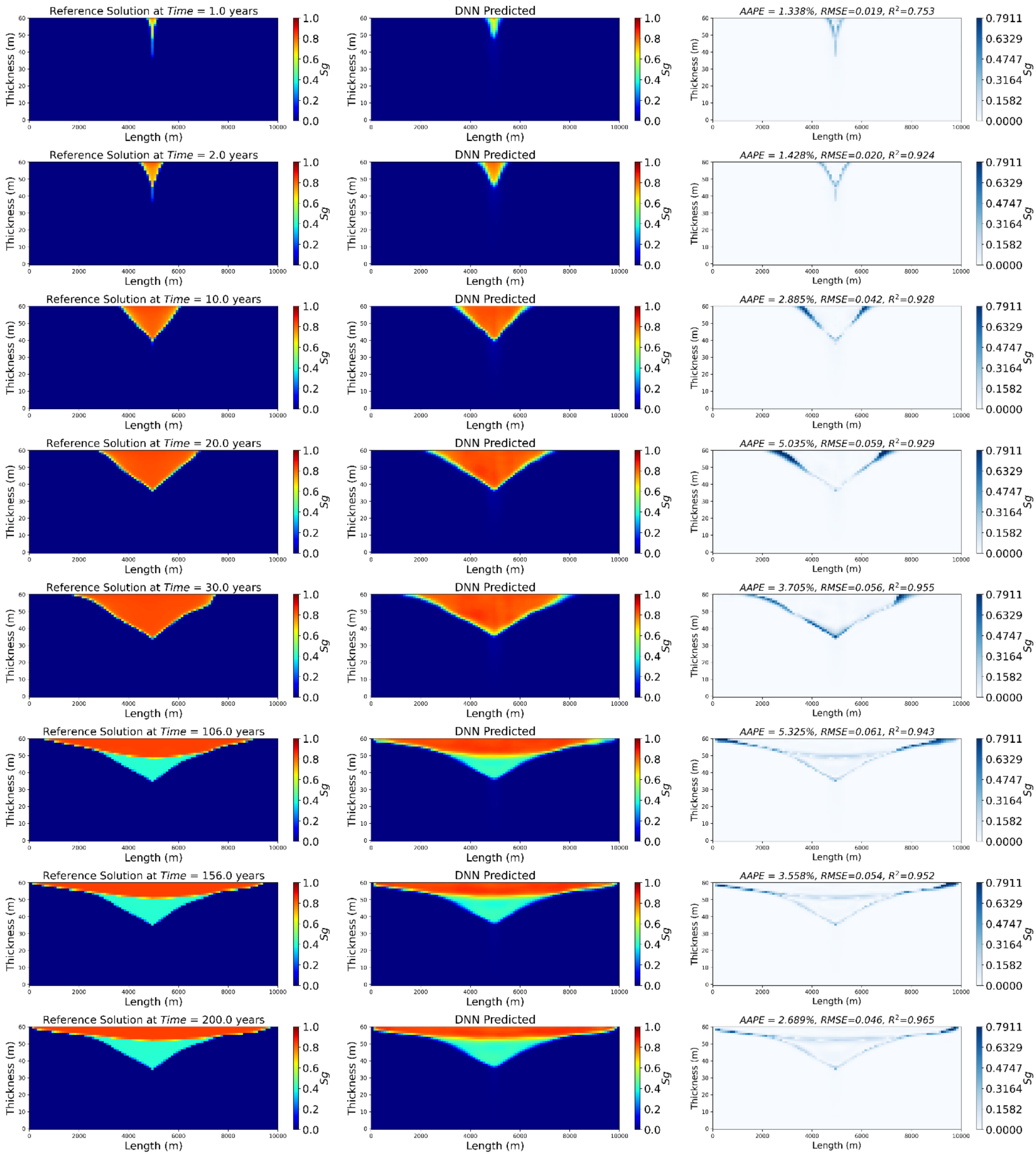


Figure 9—DL prediction, numerical simulated reference, and relative error for saturation testing case 22 at 1,2,10,20,30,100, and 200 years.

Figure 10 compares the CO_2 gas saturation predictions from the GANs model with high-fidelity numerical simulation results (CMG) for testing case number 22 at different time steps: 1, 5, 10, 30, 101, and 200 years across all grids in the x-direction at the top of the reservoir. In the early injection phase (1, 5, and 10 years), the CO_2 plume begins forming around the injection well, showing a sharp saturation peak at 1 year, which expands laterally by 5 and 10 years as the injected CO_2 spreads within the reservoir. The GANs predictions closely match the CMG reference, with minor discrepancies at the leading edge of the plume. At 30 years, marking the end of the injection phase, the CO_2 plume stabilizes with high saturation in the central region. During the post-injection period (101 and 200 years), the CO_2 plume undergoes buoyancy-driven migration and dissolution into brine, causing slight shifts in saturation distribution. While the GANs model captures the overall spatial and temporal evolution accurately, slight overestimations at the plume edges are observed in later stages. The results demonstrate that the GAN-based surrogate model provides a computationally efficient alternative to high-fidelity simulations while maintaining a high degree of accuracy.

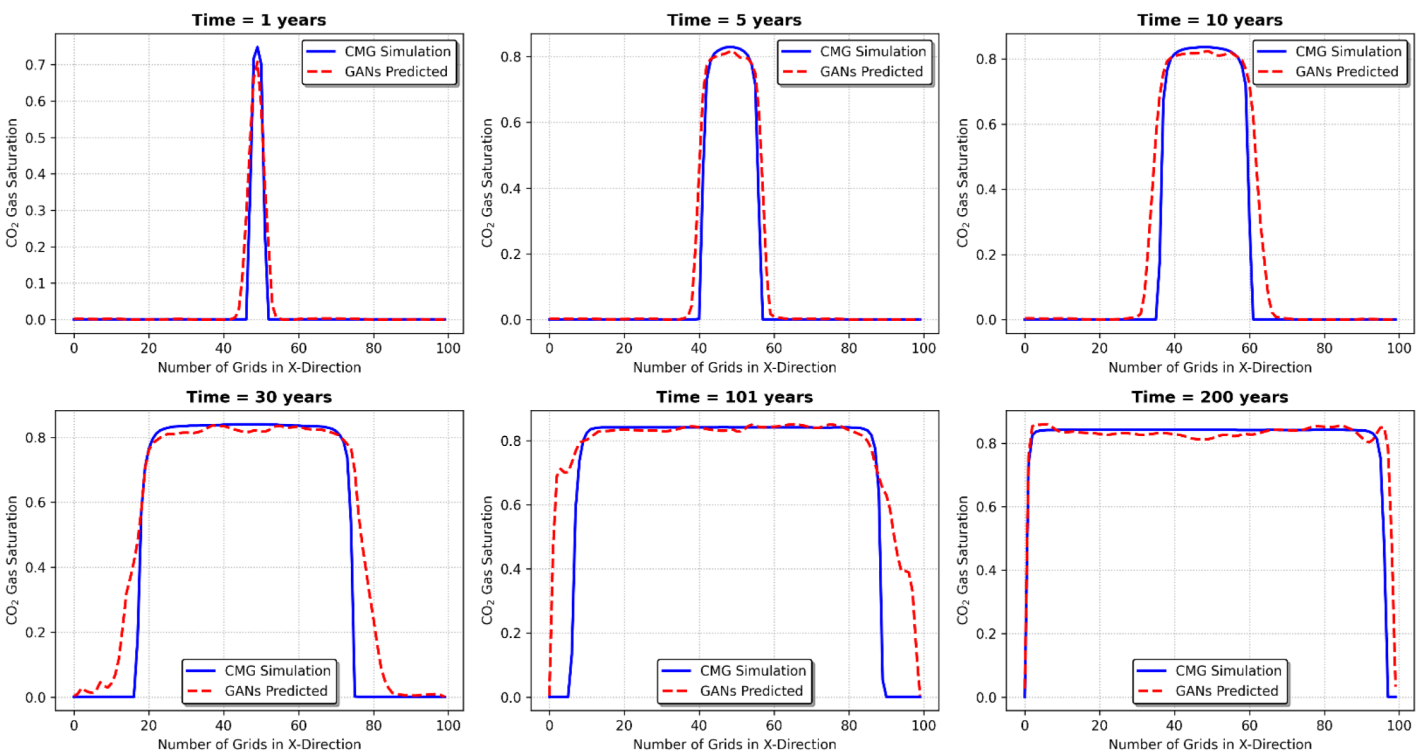


Figure 10—DL prediction and numerical simulation solution for the testing case number 22, at 1,5,10,30,100, and 200 years.

Similarly, the results for pressure buildup are illustrated in Figure 11. Unlike saturation, pressure only needs to be predicted during the injection period, as it will naturally dissipate after well shut-in, rendering post-injection monitoring unnecessary. Results at 1, 2, 4, 10, 20, and 30 years are collected for visualization. The pressure monotonically increases during injection period. The over-pressurization zone expands from a well location. A close match can also be observed between DL prediction and numerical simulation. The overall NAE is even smaller than 5%, and the average R^2 and SSIM are around 0.990. This further demonstrates the excellent performance of the proposed DL model.

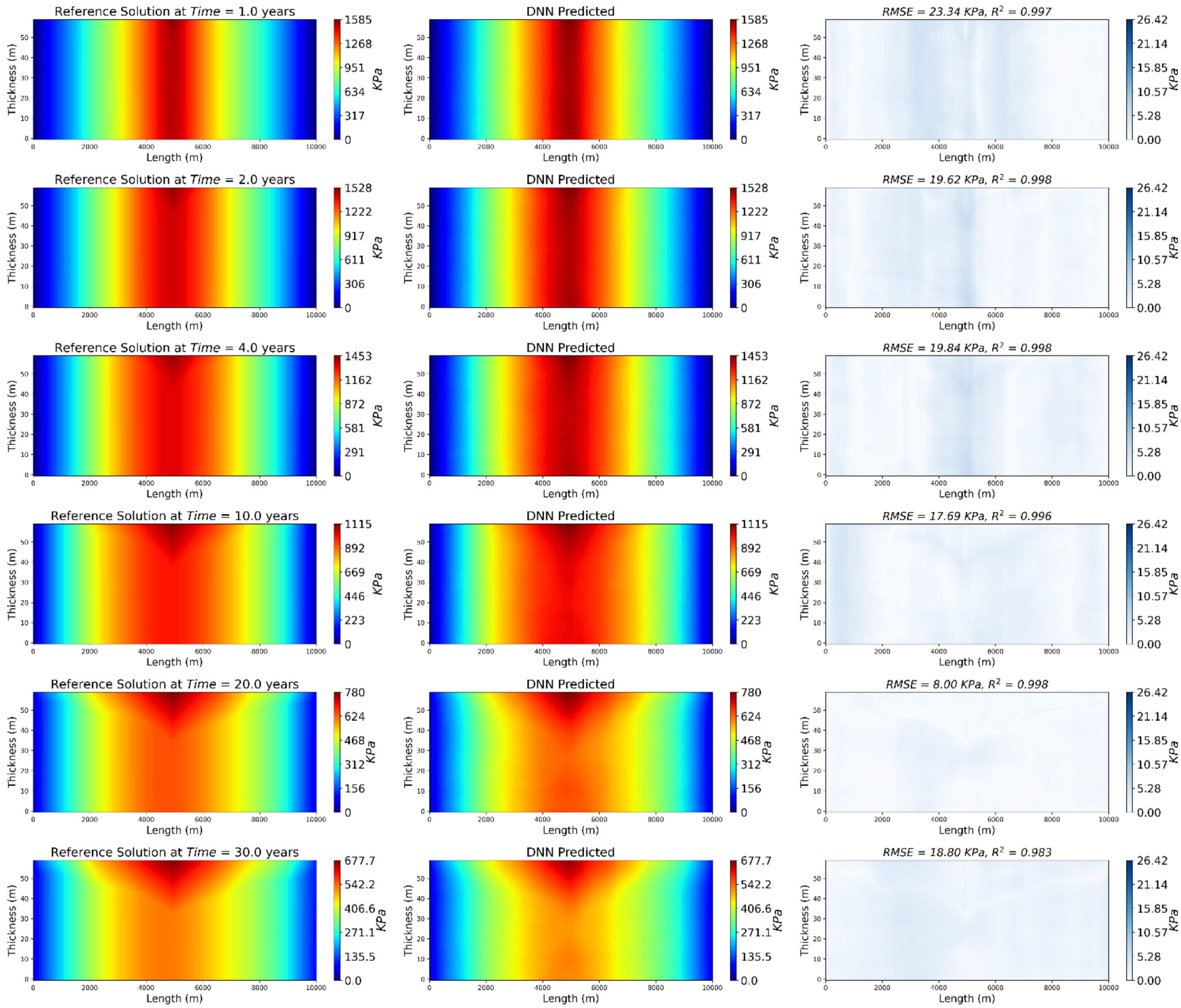


Figure 11—DL prediction, numerical simulated reference, and relative error for pressure buildup testing case number 22, at 1, 2, 4, 10, 20, and 30 years.

Figure 12 compares the CO_2 pressure buildup predictions from the GAN-based model with CMG simulation results for testing case number 22 at different time steps: 1, 2, 4, 10, 20, and 30 years, across all grids in the x-direction at the top of the reservoir. The plots illustrate how pressure distribution evolves over time due to continuous CO_2 injection. In the early stages (1 to 4 years), pressure buildup increases symmetrically around the injection well, reaching its peak at the center of the reservoir. The GANs predictions closely follow the CMG results, with minimal deviations at the peak pressure zones. By 10 years, pressure continues to increase but stabilizes as injection progresses. At 20 and 30 years, near the end of the injection period, pressure buildup reaches its maximum, forming a parabolic distribution along the x-direction. Overall, the GAN-based model effectively captures the spatial and temporal variations in pressure buildup, showing excellent agreement with CMG simulations. Minor discrepancies at later time steps suggest slight overestimations in peak pressure regions. Nevertheless, the results demonstrate that the GANs model provides a computationally efficient alternative to traditional reservoir simulations while maintaining high predictive accuracy.

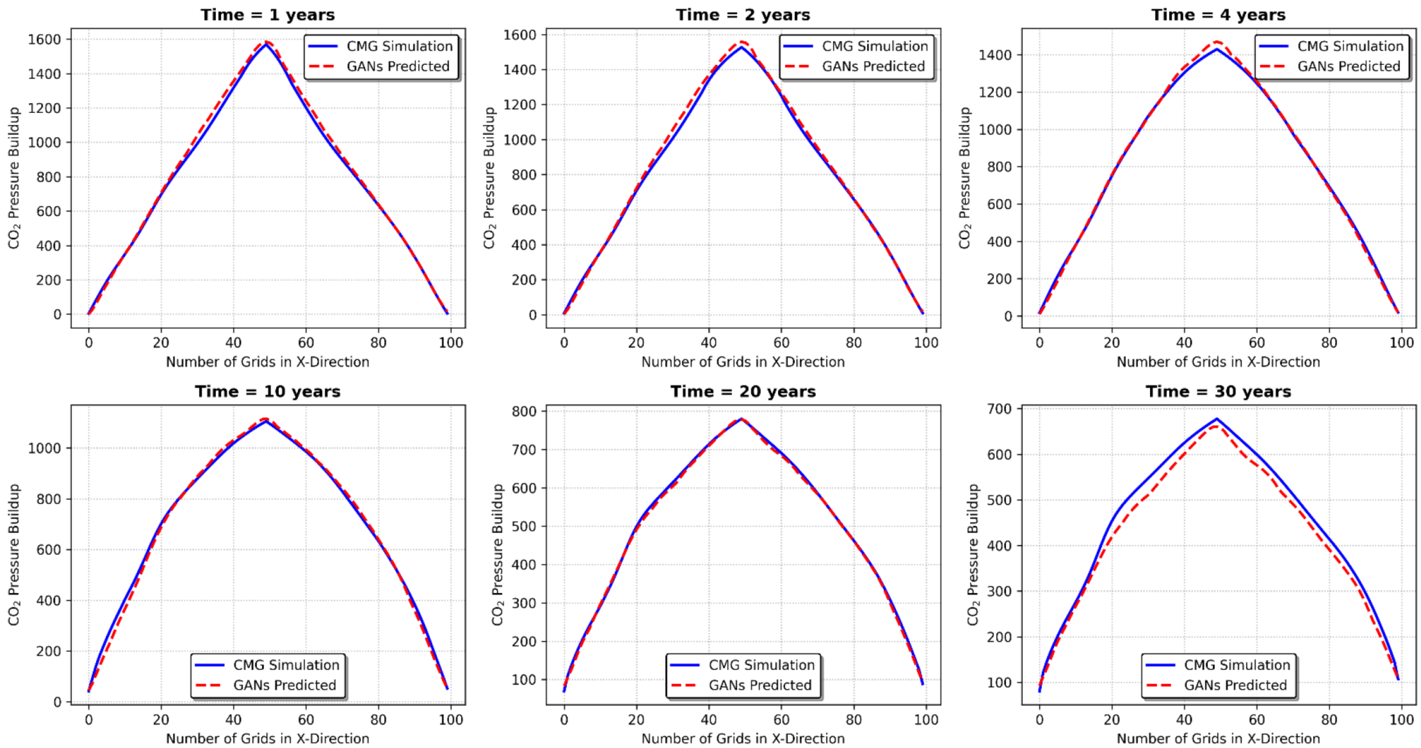


Figure 12—DL prediction and numerical simulation reference solution for predicting pressure buildup for the testing case number 22, at 1,2,4,10,20, and 30 years.

Conclusions

This study demonstrates the potential of Generative Adversarial Networks as surrogate models for predicting pressure buildup and CO_2 saturation in subsurface reservoirs. The proposed model integrated the UNet and ResNET by leveraging the local information learning capabilities of CNN and the global context modeling strengths of ResNET. Furthermore, we enhance the original convolutional layers by incorporating the multi-scale inception module. The proposed GAN-based approach significantly reduces the computational cost of traditional reservoir simulation models while maintaining high accuracy. These models can be applied in real-time CO_2 storage management systems, facilitating more efficient decision-making and optimization of CO_2 sequestration operations. Further research on improving model generalization and incorporating uncertainty quantification will enhance the applicability of GANs in large-scale CO_2 storage projects. The use of GAN-based surrogate models for predicting pressure buildup and saturation in CO_2 storage reservoirs offers several advantages. The primary benefit is the significant reduction in computational time, making it feasible to integrate these models into real-time monitoring and management systems. The GANs model provides accurate predictions without the need for exhaustive simulation runs, enabling rapid assessments of CO_2 storage performance under different conditions. However, there are challenges associated with the implementation of GANs in CO_2 storage modeling. One limitation is the need for a large, high-quality dataset to train the model effectively. Inadequate or biased data could lead to inaccurate predictions, highlighting the importance of data quality and diversity. Additionally, while GANs are powerful tools for surrogate modeling, their complexity and the need for careful tuning may make them less accessible for practitioners without deep learning expertise. Future research could explore the integration of GANs with other machine learning techniques, such as reinforcement learning, to optimize CO_2 injection strategies. Additionally, the incorporation of uncertainty quantification in GAN-based models could enhance their robustness and applicability in real-world scenarios.

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